

KAUSTAV KUNDU

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SUMMARY

Research scientist who builds and ships multimodal generative systems at scale. Curated the billion-clip datasets and captioning models behind Meta's Muse Video, a top-3 text-to-video model, after leading multimodal foundation model work at AWS that served over 2K TPS in production. Conducted doctoral research in geometry-grounded 3D scene understanding at the University of Toronto. Current focus is on closing the loop from data curation through training, evaluation, and inference for generative and embodied systems.

INDUSTRY EXPERIENCE

Staff Research Scientist, Meta (MSL)

June 2025 — Present

- Developed the ARPG-based captioning model and data-mixture ablations behind Meta's **Muse Video**, scaling the pretraining dataset from O(10M) to O(1B) clips and shipping a model ranked **#3 on the Text-to-Video Arena leaderboard**.
- Curated pretraining and mid-training datasets of O(1B) videos from a 100B+ candidate pool using distribution-aware sampling to capture long-tail concepts and improve T2V "interestingness" (human activity centric, scenic videos, etc.).
- Built a vision-language agent for long-horizon embodied procedural tasks using subgoal decomposition + self-correction; improved task success and recovery in failure cases via state-consistency checks and iterative replanning, released as an arXiv preprint under review at **NeurIPS 2026**.

Senior Applied Scientist, AWS AI

October 2019 — June 2025

- Led a cross-team research effort to train a collection of **3B to 34B multimodal models** on visual, text, and audio input modalities for **fine-grained reasoning with reduced hallucination**.
 - Instruction tuning, preference learning for multi-turn conversations
- *Tech lead (TL)* of a research team of 7 scientists which deployed 3 Moderation API launches in 2.5 years which had **>2K TPS** traffic and **seven-digit annual revenue**.
 - Built the research agenda to achieve **112x FPR reduction** and **41% FNR reduction**.
 - *Open set image recognition*: Generalization preserving adaptation of FMs, Backward compatibility.
 - *Self supervised learning*: Long form video understanding, Person tracking
- Led research projects that produced **8 publications (3 orals)** in CVPR, ECCV, NeurIPS with an **acceptance record of 80%**.
- Mentored 9 full-time research scientists and 9 PhD interns.

Applied Scientist, Amazon Go

March 2018 — October 2019

- Led the research and development of the **first end-to-end computer vision based real-time action detection system** in an Amazon Go store. This enabled the stores to sell non-discrete items (such as beverages, hot foods) in the Amazon Go stores. Was deployed from 2019 to (at least) 2022 in more than 10 1P stores across North America and Europe.
- Designed and implemented a multi-task based real-time deep learning model which enabled the Amazon Go stores to remove depth sensors and **reduce CapEx by 50% without loss in accuracy**.

Research Intern, Apple (SPG)

June 2016 — September 2016

- Designed the research direction and implemented deep structured Models to predict lane boundaries in autonomus driving scenes which improved F1 score on actual datasets by 48%.

Research Intern, Apple (SPG)

May 2015 — August 2015

- Implemented a CNN based deep learning network to predict the future 0.5s trajectory of each vehicle in the autonomus driving scenes. Explored use of simulation data and homography transformation to reason in the BEV space.

EDUCATION

PhD (ABD) in Computer Science, University of Toronto

January 2014 — December 2017

Thesis Title: Efficient Search Strategies in 3D for Visual Scene Understanding.

Advisors: Raquel Urtasun and Sanja Fidler

Masters in Computer Science, Toyota Technological Institute at Chicago

September 2012 — December 2013

Thesis Title: Joint Semantic Segmentation and Depth Prediction in 3D Point Cloud.

Advisor: Raquel Urtasun

BTech (Hons.) in Computer Science and Engineering, IIIT Hyderabad

August 2008 — May 2012

Thesis Title: Geometry directed browser for personal photographs.

Advisor: P J Narayanan

PUBLICATIONS

Under Review

- *Plan, Watch, Recover: A Benchmark and Architectures for Proactive Procedural Assistance*. [Kaustav Kundu](#), Ritvik Shrivastava, Maxim Arap, Nanshu Wang, Xianhui Zhu, Quintin Fettes, Gautam Tiwari, Parth Suresh, Théo Moutakanni, Alejandro Castillejo Munoz, Allen Bolourchi, Pascale Fung, Pinar Donmez, Babak Damavandi, Anuj Kumar, Seungwhan Moon. **NeurIPS 2026**

Peer Reviewed

- *Robustness Preserving Fine-tuning using Neuron Importance*. Guangrui Li, Rahul Duggal, Aaditya Singh, Bing Shuai, [Kaustav Kundu](#), Jon Wu. **ECCV 2024**
- *Towards Omnisupervised Instance Segmentation with Foundation Models*. Arnav Das, Ritwick Chaudhry, [Kaustav Kundu](#), Davide Modolo. **CVPRW 2024**
- *Hierarchical Self-supervised Representation Learning for Movie Understanding*. Fanyi Xiao, [Kaustav Kundu](#), Joseph Tighe, Davide Modolo. **CVPR 2022**
- *Id-Free Person Similarity Learning*. Bing Shuai, Xinyu Li, [Kaustav Kundu](#), Joseph Tighe. **CVPR 2022**
- *What to Look at and Where: Semantic and Spatial Refined Transformer for Detecting Human-Object Interactions*. ASM Iftekhar, Hao Chen, [Kaustav Kundu](#), Xinyu Li, Joseph Tighe, Davide Modolo. **CVPR 2022 (Oral)**
- *TubeR: Tubelet Transformer for Video Action Detection*. Zhao et al. **CVPR 2022 (Oral)**
- *Positive-congruent training: Towards regression-free model updates*. Sijie Yan, Yuanjun Xiong, [Kaustav Kundu](#), Shuo Yang, Siqi Deng, Meng Wang, Wei Xia, Stefano Soatto. **CVPR 2021 (Oral)**
- *Exploiting weakly supervised visual patterns to learn from partial annotations*. [Kaustav Kundu](#), Erhan Bas, Michael Lam, Hao Chen, Davide Modolo, Joseph Tighe. **NeurIPS 2020**
- *Pose Estimation for Objects with Rotational Symmetry*. Enric Corona, [Kaustav Kundu](#), Sanja Fidler. **IROS 2018**
- *SurfConv: Bridging 3D and 2D Convolution for RGBD Images*. Hang Chu, Wei-Chiu Ma, [Kaustav Kundu](#), Raquel Urtasun, Sanja Fidler. **CVPR 2018**
- *3D Object Proposals using Stereo Imagery for Accurate Object Class Detection*. Xiaozhi Chen*, [Kaustav Kundu](#)*, Yukun Zhu, Humin Ma, Sanja Fidler, Raquel Urtasun. **TPAMI 2017**
- *Annotating Object Instances with a Polygon-RNN*. Lluís Castrejón, [Kaustav Kundu](#), Raquel Urtasun, Sanja Fidler. **CVPR 2017 (Best Paper Honorable Mention Award)**
- *Exploiting Semantic Information and Deep Matching for Optical Flow*. Min Bai*, Wenjie Luo*, [Kaustav Kundu](#), Raquel Urtasun. **ECCV 2016**
- *Monocular 3D Object Detection for Autonomous Driving*. Xiaozhi Chen, [Kaustav Kundu](#), Ziyu Zhang, Humin Ma, Sanja Fidler, Raquel Urtasun. **CVPR 2016**
- *3D Object Proposals for Accurate Object Class Detection*. Xiaozhi Chen*, [Kaustav Kundu](#)*, Yukun Zhu, Andrew Berneshawi, Humin Ma, Sanja Fidler, Raquel Urtasun. **NeurIPS 2015**
- *Rent3D: Floor-Plan Priors for Monocular Layout Estimation*. Chenxi Liu*, Alexander Schwing*, [Kaustav Kundu](#), Raquel Urtasun, Sanja Fidler. **CVPR 2015 (Oral)**
- *Geometry Directed Browser For Personal Photographs*. Aditya Deshpande, Siddharth Choudhary, P J Narayanan, Krishna Kumar Singh, [Kaustav Kundu](#), Aditya Singh, Apurva Kumar. **ICVGIP 2012 (Oral)**

SKILLS

Distributed training	NCCL, expert parallelism for MoE, DeepSpeed ZeRO 3, FSDP, multi-rank coordination at 64 node scale
RL for LLMs	GRPO, DAPO, DPO, PPO and variants, REINFORCE, advantage regularized PG
Multimodal	VLM, MLLM, generative modeling, spatiotemporal reasoning, grounding, hallucination mitigation
Data & Eval	Curation at O(1B) scale, synthetic data, expert-model filtering, benchmarking, error analysis
Frameworks	PyTorch, vLLM, LLaMA-Factory, WebDataset, FiftyOne
Production ML	2K+ TPS serving, real-time inference, monitoring and reliability

AWARDS

- Best Paper Honorable Mention, **CVPR 2017**
- Outstanding Reviewer, **CVPR 2018**, **CVPR 2021**

ACTIVITIES

Tutorial/Workshop Organizer

Wearable AI Workshop at **ECCV 2026**.

Workshop on Multimodal Content Moderation at **CVPR 2024**.

Cross-Model Compatibility in Computer Vision at **ICCV 2021**.

Program Committee

Area Chair: **CVPR 2026**, **WACV 2027**.

Conferences: CVPR 2018 to 2026, ECCV 2018 to 2026, ICCV 2019 to 2023, ICLR 2021 to 2025, NeurIPS 2020 to 2026, ICML 2022 to 2025

Journals: T-PAMI 2018 to present